TEST REPORT FORM

Procedure used: ☒ 3E **VERSION DATE: 2026**

> ISTA CERTIFIED LABORATORY

Laboratory: **Zhongbao Test Technology (Wuxi) Co., Ltd.**

Address: **No. 39, Hanlin Road, Ehu Town, Xishan District, Wuxi, Jiangsu, China**

City: **Wuxi** State/Prov: **Jiangsu**

Zip/Postal Code: **214117** Country: **China**

ISTA Member ID: **11370**

Technician Performing Test: **Jiao Qingyu**

Email: **jiaoqingyu@packagetest.net**

Lab reference number for test *(if applicable)*: **2026-Y009-211**

> PRODUCT MANUFACTURER / SHIPPER

Test Requested By: **Wang Hongmei**

Company: **Qingdao Tengcheng Packaging Co., Ltd.**

Address: **No. 497, Huaishe River 1st Road, Wangjiayuan Village, Tongji Sub-district, Jimo District, Qingdao City, Shandong Province**

City: **Qingdao** State/Prov.: **Shandong**

Zip/Postal Code: **/** Country: **China**

Phone: **+86-15866807268**

Email: **zhouhai_lu@126.com**

ISTA Member? ☐ Yes ☒ No

ISTA Member ID *(if applicable)*: **/**

Manufacturer's License Number *(if applicable)*: **/**

> TEST DETAILS ALL FIELDS IN THIS SECTION MUST BE COMPLETED

Date Tested: **May.14, 2026 - May.14, 2026**

Number of samples tested: **1**

Number of replicate tests performed: **0**

Gross Weight: **1000.0 kg**

External Container Size (LxWxH): **1100 mm×1100 mm×1165 mm**

Product Damage Tolerance (PDT): **/**
Package Degradation Allowance (PDA): **The appearance of packaged-product was normal.**
PDT/PDA Determined By/Date: **Wang Hongmei / May.7, 2026**

Method used to determine Pass/Fail: **Visual check**

Person determining Pass/Fail result: **Wang Dongfang**

> PRODUCT AND PACKAGE DESCRIPTIONS ALL FIELDS IN THIS SECTION MUST BE COMPLETED

It is strongly recommended that photographs, detailed drawings, and/or complete specifications of product and exterior and interior packaging accompany this report. If there is insufficient information supplied by the product manufacturer, please indicate the reason in the TEST RESULTS section, at the end of this report form.

Specific PRODUCT TESTED: Include, as applicable, product name, brand, model number, serial number and similar information that will help to identify the specific product tested.
Sample Name: PAPER IBC Packaged-Products
Model: 1000L 8ply

PRODUCT Description: Describe product in detail. Include type of product, accessories and other identifying information, including specifics on bottles, containers and liquid or solid contents.

Did the lab OPEN the packaged-product before testing to determine product condition? ☐ YES ☒ NO

If YES to above, list PRODUCT Condition before testing:
List any damage or irregularities seen prior to testing.

Was the PACKAGING used during testing:
☒ **ORIGINAL** as arrived in the lab for testing
☐ **NEW** re-packaged with new materials before testing

PACKAGE Condition before testing: List any damage or irregularities seen prior to testing.
The appearance of packaged-product was normal.

PACKAGE Description: Describe entire shipping unit. Description must be detailed and specific and should include type, style and material of packaging; corrugated board composition; cushion details including performance; film gauge and composition; application or package forming details; mold numbers; any pallet or skid; unitization method for unit loads; methods of closure, etc.
Pallet packaged-product, with four-way forklift available pallet at the bottom, secured by 4 plastic bands, printing on the surface. The appearance of the packaged-product is normal.

ista TEST METHODS PROCEDURE 3E

> ORIENTATION

What position was the packaged-product in when the faces, edges and corners were identified:

☐ **MOST STABLE** ☒ **SHIPPING**

Explain position if different than procedure recommendation: **/**

> ATMOSPHERIC CONDITIONING

Required Preconditioning (Ambient)

Time of Conditioning (hours): **12** Temperature (°F / °C): **28℃~28℃** Humidity (%): **60~63**

Start of test: Temperature (°F / °C): **28℃** Humidity (%): **60**

End of test: Temperature (°F / °C): **28℃** Humidity (%): **60**

Optional Conditioning (Controlled)

Time of Conditioning (hours): **/** Temperature (°F / °C): **/** Humidity (%): **/**

> SHOCK TEST

Method Used: ☒ **Incline-Impact** ☐ **Horizontal Sled**

Sequence Number	Velocity (fps)	Orientation of packaged-product (ex: Face 6; Edge 3-5)	
1	**1.2m/s**	**FACE**	**5**
2	**1.2m/s**	**FACE**	**6**
3	**1.2m/s**	**FACE**	**2**
4	**1.2m/s**	**FACE**	**4**
Rotational Edge Drop #1	**150mm**	**EDGE**	**3-6**
Rotational Edge Drop #2	**150mm**	**EDGE**	**2-3**

> COMPRESSION TEST

Method Used: ☒ **Machine** ☐ **Dead Weights and Load Spreader**

Formula, with values, used to calculate Test Load: **Wt×(S-1)×F×9.8×C**

=1000.0×(3-1)×3×9.8×1.4=82320 N

Compensating Factor **(F)** used: **F=3**

If the Compensating Factor used is NOT ISTA recommended, please explain: **/**

Warehouse Stack **(S)** used, if applicable: **S=3**

Machine:

Calculated Test Load *(lbf / N)*: **82320 N** Actual Test Load *(lbf / N)*: **82320 N** Total Test Time (minutes): **/**

Dead Weights and Load Spreader:

Calculated Test Load *(lbs/kg)*: **/** Actual Test Load *(lbs/kg)*: **/** Total Test Time (minutes): **/**

> VIBRATION TEST

Packaged-Product Orientation during test: **Transport orientation**

Describe restraining devices used, if any: **Aluminum fense**

Anticipated Transport Miles: **/** Calculated Test Time in minutes: **240**

Use the spaces below to record frequency and PSD levels used for random vibration:

Frequency (Hz)	PSD Level, g^2/Hz
1.0	0.00072
3.0	0.018
4.0	0.018
6.0	0.00072
12.0	0.00072
16.0	0.0036
25.0	0.0036
30.0	0.00072
40.0	0.0036
80.0	0.0036
100.0	0.00036
200.0	0.000018
Overall Grms:	0.54

> SHOCK TEST

Use the spaces below to record drop heights and orientations of each drop:

Sequence Number	Height	Orientation (ex: Edge 3-5)	
Rotational Edge Drop #1	150mm	EDGE	3-5
Rotational Edge Drop #2	150mm	EDGE	3-4

> TEST RESULTS

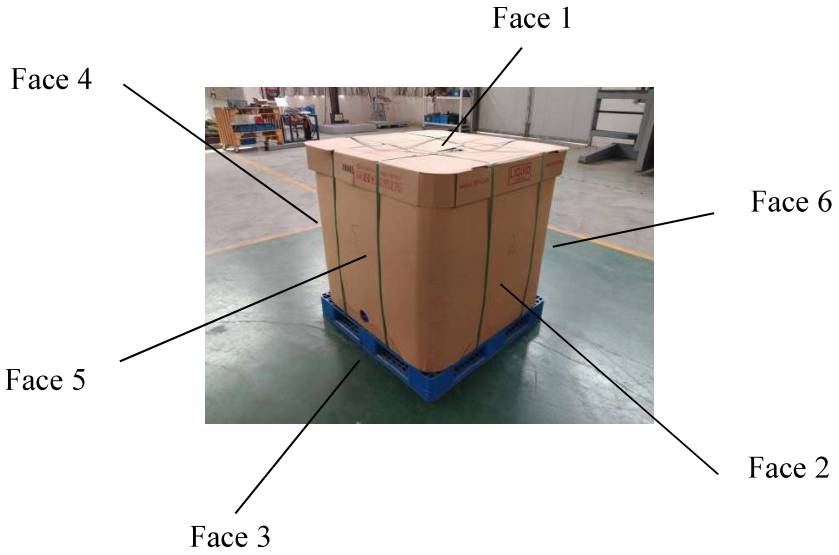
PRODUCT Condition after testing (if inspected)**: /**

PACKAGE Condition after testing: The appearance of packaged-product was normal.

☒ **Pass** ☐ **Fail**

Comments or recommendations:
/

Appearance Check & Direction Pictures:



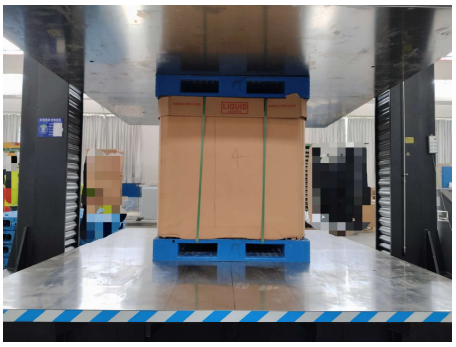
Test Pictures:



Incline impact test



Rotational edge drop test



Compression Test



Random vibration test



Rotational edge drop test

Post Test Pictures:








The appearance of packaged-product was normal.